Is Cholesterol Overrated? Examining Its Independent Effect on Heart Disease Risk

Logan Calder, Chase Mayer, Anirudh Ramesh

Research Question

- **In this project, we are investigating…**
  - What is the relationship between serum cholesterol (mg/dL) and the presence of coronary heart disease, and how does that relationship change after adjusting for age and resting blood pressure?
- **Why it matters**
  - Cholesterol drives decades of dietary guidelines and pharmaceutical development
  - Recent evidence suggests its independent signal weakens once age and BP are controlled for
  - This is a purely **descriptive** question — not causal inference

Operationalization and Variables

- **Outcome (Y):** Binary heart disease indicator — 1 if vessel narrowing present, 0 otherwise
- **Primary predictor (X):** Serum cholesterol (mg/dL) from fasting blood draw
- **Covariates:** Age (years), Resting blood pressure (mmHg)
- **Slippage to consider**
  - Binary outcome discards disease severity (original scale: 0–4)
  - Cholesterol is a single snapshot — does not capture trends over time
  - Statin use not recorded — could suppress measured cholesterol in high-risk patients

Population

- **Dataset:** UCI Heart Disease dataset (Janosi et al., 1988) — 4 clinical sites
- **Unit of observation:** Individual patient visit with angiographic examination
- **Sample construction**
  - Primary: Cleveland Clinic (n = 304 before cleaning, 303 after); pooled four-site (n = 675) for replication
  - Cleveland split 70/30: 70% confirmation set (n = 213), 30% exploration set
  - Clinical convenience sample — higher-risk patients referred for angiography
  - Base rate of heart disease ~47% — far higher than general population; do not extrapolate

Regression Results

- **4 Linear Probability Models:** Cleveland + all-sites × unadjusted + adjusted. HC3 robust SEs throughout.
- **Key findings**
  - Cholesterol not significant in Cleveland alone ($p \approx 0.26$ unadjusted; $p \approx 0.72$ adjusted)
  - Cholesterol reaches significance in all-sites (larger $n = 675$), but effect size identical: +100 mg/dL → ~10 pp increase
  - Age is the dominant predictor: each additional decade → ~9 pp ($p < 0.05$)
  - Model fit: adjusted $R^2 = 0.055$ (Cleveland), 0.093 (all sites) — low in absolute terms
- **Practical significance (Model 2, sample medians: chol=243, age=56, BP=130)**
  - A 100 mg/dL increase in Cholesterol shifts predicted probability by ~10pp
  - Age 56→66 (same cholesterol) shifts predicted probability by ~9 pp

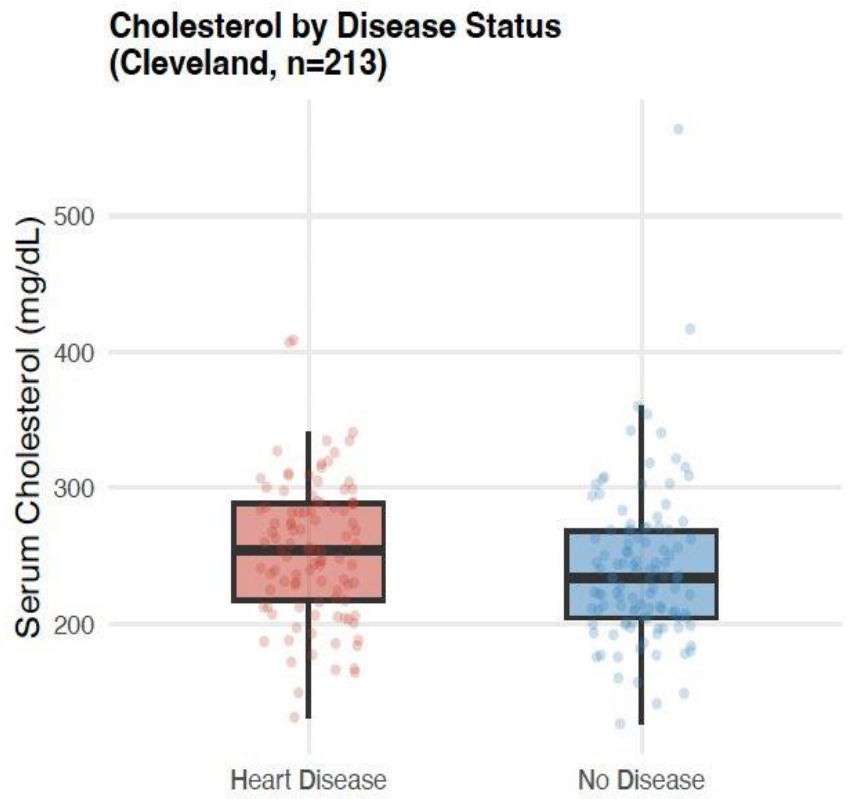
Regression Results

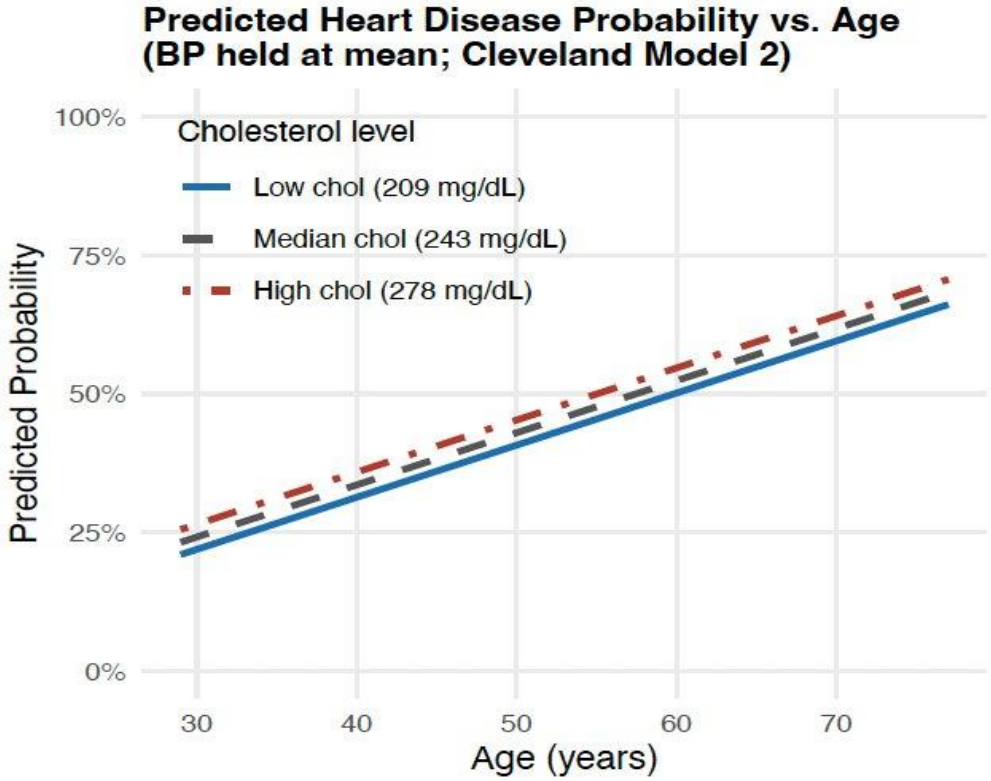
Table 1: Serum Cholesterol and Heart Disease Presence

	Dependent variable:			
	Heart Disease (1 = present, 0 = absent)			
	Cleveland Only		All Four Sites	
	(1)	(2)	(3)	(4)
Serum Cholesterol (mg/dL)	0.001	0.001	0.001***	0.001**
	(0.001)	(0.001)	(0.0003)	(0.0003)
Age (years)		0.009**		0.013***
		(0.004)		(0.002)
Resting Blood Pressure (mmHg)		0.004*		0.003***
		(0.002)		(0.001)
Constant	0.191	−0.671**	0.224***	−0.779***
	(0.188)	(0.293)	(0.082)	(0.166)
Observations	213	213	675	675
R^2	0.016	0.068	0.014	0.097
Adjusted R^2	0.011	0.055	0.012	0.093
Residual Std. Error	0.498 (df = 211)	0.487 (df = 209)	0.497 (df = 673)	0.476 (df = 671)
F Statistic	3.415* (df = 1; 211)	5.116*** (df = 3; 209)	9.473*** (df = 1; 673)	23.928*** (df = 3; 671)

Note: *p<0.1; **p<0.05; ***p<0.01
Robust (HC3) standard errors in parentheses.

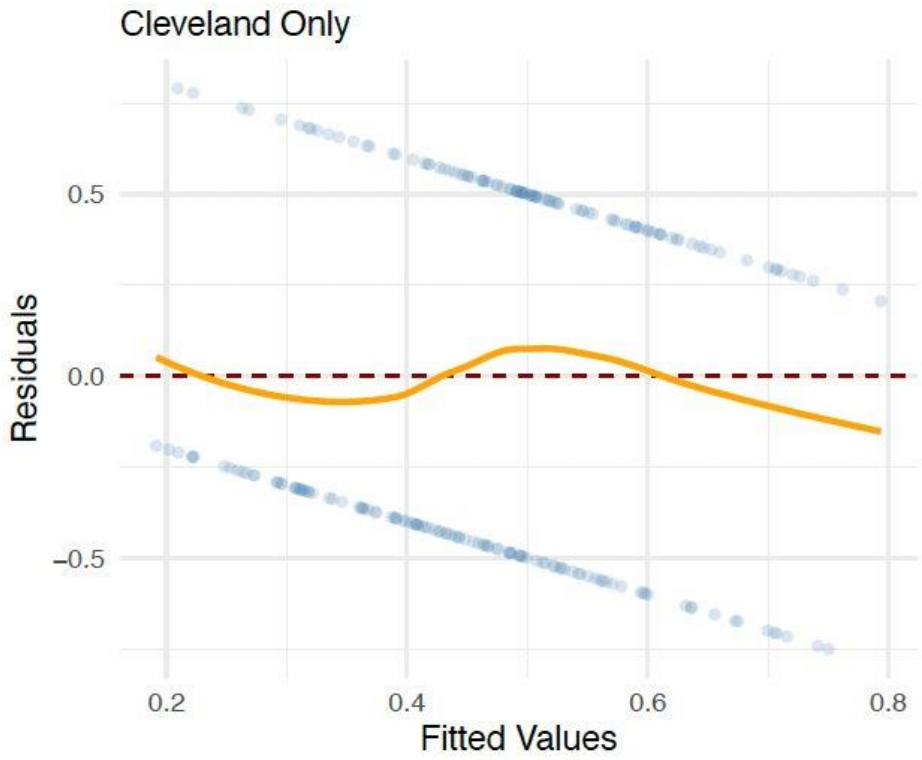
Data Visualization: What the Data Shows

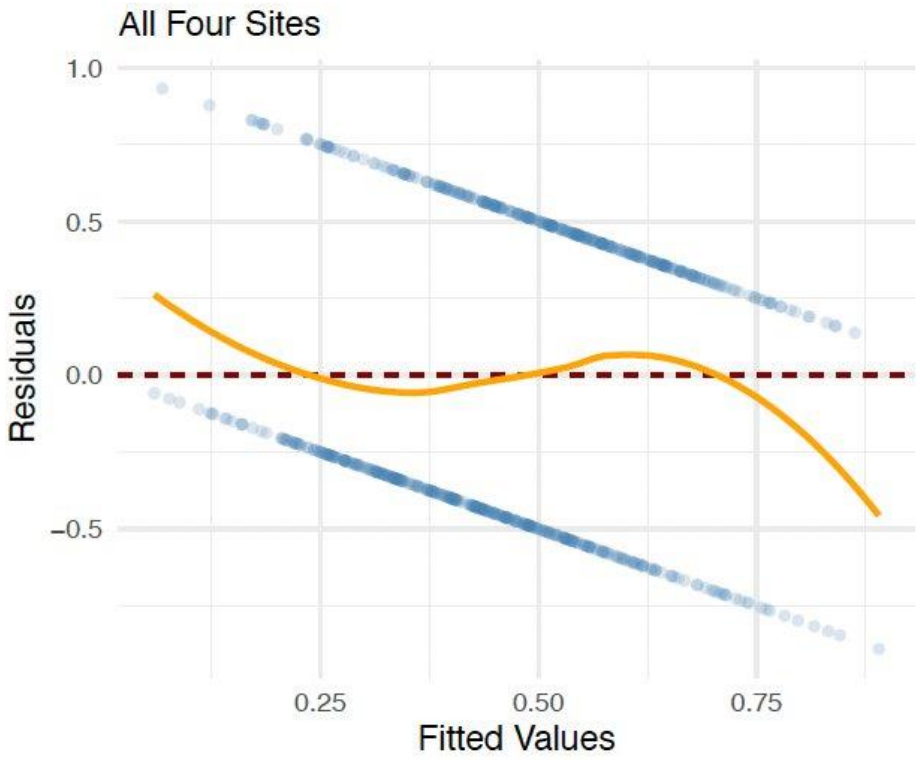




Left: Cholesterol IQRs overlap almost entirely — no visual separation between disease and no-disease groups
Right: Three cholesterol lines (low/median/high) track nearly on top of each other; age 30→70 drives ~40pp increase in predicted risk

Residuals vs. Fitted: Assumption Checks





- Loess smoother tracks near zero — no systematic curvature, linear CEF supported
- Pairwise correlations $|r| < 0.35$; VIF < 1.15 — no multicollinearity concern
- Heteroskedasticity expected with binary outcome — addressed throughout with HC3 robust SEs

Conclusion

- Cholesterol shows a **statistically fragile and practically small** association with heart disease
- Its descriptive signal is largely absorbed by **age** and, to a lesser extent, **blood pressure**
- **Key limitation:** Statin use not recorded — could suppress cholesterol in highest-risk patients, potentially understating its relevance
- Questions for the audience:
  - "We used a linear probability model rather than logistic regression for interpretability — is that the right tradeoff here?"
  - "Are there other confounders beyond age and BP we should have included?"
  - "We excluded observations where cholesterol was recorded as zero, treating them as physiologically implausible rather than true zero values. Could this exclusion have introduced selection bias — and if so, in which direction?"
  - "Our sample consists entirely of patients already referred for angiography, meaning they were already flagged as higher-risk. How do you think this referral bias affects the generalizability of our cholesterol finding to the broader population?"

References

- Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1988). *Heart Disease* [Dataset]. UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/45/heart+disease
- Wilson, P. W., et al. (1998). Prediction of coronary heart disease using risk factor categories. *Circulation*. https://www.ahajournals.org/doi/epub/10.1161/01.CIR.97.18.1837
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